TEB3133/TFB3133 Data Visualization
Project

Due Date: 20 July 2026

Stakeholder Request:

As a Business Intelligence Manager, I oversee large-scale multivariate datasets generated from multiple departments. Senior management requires an interactive dashboard that can transform complex datasets into actionable insights for strategic decision-making.

Your team has been assigned to design and develop a professional dashboard solution that supports different user groups, follows visualization standards, and presents meaningful insights from the data.

The final dashboard should not only display information but also enable users to explore patterns, trends, anomalies, and business opportunities through interactive visual analytics.

Requirements

Task 1: Select a Domain

Choose ONE of the following domains:

- Healthcare
- Education
- E-Commerce
- Industrial Manufacturing
- Oil & Gas
- Finance
- Entertainment

Task 2: User-Centered Design

Apply visualization design principles for at least THREE user categories:

User Group	Design Considerations
Early Childhood	Large icons, minimal text, bright colors, simple navigation
Adults	Information-rich dashboards, filtering, comparative analysis
Elderly Users	High contrast, larger fonts, simplified interactions, accessibility support

Your report should justify how the design adapts to each user group.

Task 3: Dashboard Development

Develop the dashboard using any ONE of the following tools:

- D3.js
- Tableau
- Power BI
- Grafana
- Other approved visualization platform

The dashboard must contain:

- KPIs and Summary Cards
- Interactive Filters
- Drill-down Analysis
- Dynamic Charts
- Responsive Layout
- User-friendly Navigation

Task 4: Interactive Features

Implement at least FOUR interactive features:

Examples:

- Filtering
- Zooming
- Panning
- Highlighting
- Tooltip Information
- Cross-filtering
- Dynamic Search
- Drill-down Analysis
- Dashboard Linking

All interactions should follow usability and accessibility standards.

Task 5: Advanced Visualization Techniques

Select and implement at least TWO advanced visualization techniques:

- Heat Map
- Parallel Coordinates Plot
- Trellis Plot
- Mosaic Plot
- Scatter Plot Matrix
- Grand Tour Visualization

For each technique:

1. Explain why it was selected.
2. Describe what insights it reveals.
3. Compare it against a traditional chart.

Task 6: Timeline Visualization

Using a timeline library or custom implementation:

Examples:

- TimelineJS
- Vis.js Timeline
- D3 Timeline
- JavaScript Timeline Components

Create a project progress timeline covering the last **5 months**.

The timeline should include:

- Project Initiation
- Data Collection
- Data Cleaning
- Dashboard Development
- Testing and Validation
- Deployment/Presentation

The timeline must be interactive and visually appealing.

Task 7: Insight Generation

The dashboard should go beyond reporting and provide actionable insights.

Examples:

Healthcare

- Disease outbreak hotspots
- High-risk patient groups
- Resource utilization trends

Education

- Student performance trends
- Attendance-risk identification
- Learning outcome comparison

E-Commerce

- Product demand forecasting
- Customer segmentation
- Regional sales analysis

Oil & Gas

- Equipment failure prediction
- Production efficiency analysis
- Safety incident monitoring

Finance

- Fraud detection indicators
- Investment performance trends
- Customer spending patterns

Deliverables

1. Dashboard Application
2. Source Code / Project Files
3. Dataset
4. Project Report (5–10 pages)
5. Poster (Soft copy)
6. Demonstration (During Class time in Week 11)

Evaluation Criteria (100 Marks)

Criteria	Weight (%)	5 - Excellent	4 - Good	3 - Satisfactory	2 - Fair	1 - Poor
Domain Selection & Problem Definition	10	Clearly defines domain, business problem, objectives, and stakeholders with strong justification.	Problem and objectives are clear with minor gaps.	Problem is identified but lacks sufficient detail or justification.	Problem definition is unclear and objectives are weak.	Domain selection and problem statement are missing or inappropriate.
User-Centered Design (Child, Adult, Elderly)	15	Design fully accommodates all three user groups with excellent usability and accessibility considerations.	Design supports most user groups with good usability.	Basic consideration of user groups but limited adaptation.	Minimal user-centered design principles applied.	No evidence of user-centered design.
Dashboard Functionality	15	Dashboard is fully functional, responsive, professional, and error-free.	Dashboard functions well with minor issues.	Dashboard meets basic requirements but lacks polish.	Dashboard has several functionality issues.	Dashboard is incomplete or non-functional.
Interactivity Implementation	15	Multiple meaningful interactive features implemented effectively and enhance analysis.	Good use of interactivity with minor limitations.	Basic interactive features implemented.	Limited interactivity with minimal value.	No interactive features implemented.
Advanced Visualization Techniques	15	Two or more advanced visualizations are correctly implemented and generate valuable insights.	Advanced visualizations are implemented with good explanation.	Visualizations are implemented but insights are limited.	Visualizations are poorly selected or weakly explained.	Advanced visualization requirements are not met.
Timeline Visualization	10	Interactive timeline is highly informative, visually appealing, and professionally designed.	Timeline is complete and easy to understand.	Timeline includes major milestones but lacks detail.	Timeline is incomplete or poorly designed.	Timeline is missing.
Data Storytelling & Insights	10	Excellent interpretation with actionable insights and recommendations.	Good insights with clear interpretation.	Basic insights provided with limited analysis.	Minimal interpretation of results.	No meaningful insights or conclusions.
Technical Implementation	5	Clean, efficient, well-organized implementation with proper documentation.	Good implementation with minor issues.	Functional implementation with average organization.	Poor organization and technical quality.	Incomplete implementation or major technical issues.
Presentation & Demonstration	5	Professional presentation, clear communication, and excellent demonstration.	Good presentation with clear explanation.	Adequate presentation and demonstration.	Weak presentation with limited explanation.	Poor or missing presentation.